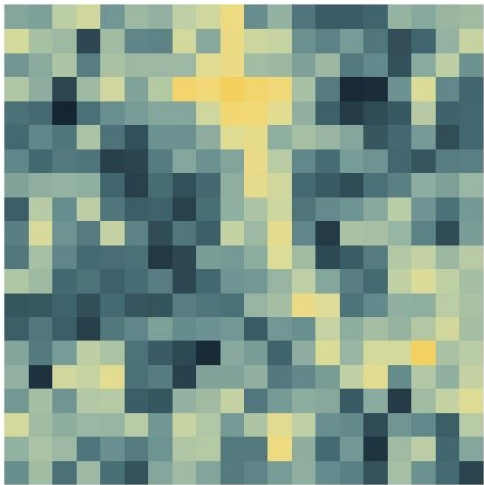


THE GAUSSIAN FREE FIELD

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This is an introduction to the Gaussian free field and its relations to conformal field theory.



1. INTRODUCTION

1.1. **Later added parts.** Gaussian random chaos corresponds to the Virasoro (and is related to Liouville quantum gravity). The Toda field theories correspond to affine W algebras, this is recent work.

CFT	Prob	VA
rank one boson	GFF	Heiseneberg
Liouville quantum gravity	Gaussian random chaos	Virasoro
?	Toda field theories	W algebras

1.2. **Non-definition.** The Gaussian free field is morally a random variable

$$\psi(x)$$

attached to every point x in an open set $D \subseteq \mathbf{R}^d$. Each of these is Gaussian of mean zero and the covariance

$$\mathbf{E}(\psi(x)\psi(y)) = G_D(x,y)$$

Date: February 24, 2022.

is the Green's function for the Laplacian. It is important because in two dimensions, this Green's function is $\log(x - y)$, and so taking derivatives we get the genus zero correlation function of the rank one free boson [FB]:

$$\mathbf{E}(\psi'(z)\psi'(w)) = \frac{1}{(z - w)^2}.$$

1.3. Definition. The above definition does not quite work because of the singularities in $G_D(x, y)$. To fix this we “blur” the points, and instead of actual points $x \in D$ consider measures $\mu \in \mathcal{M}$; here $\mathcal{M}^+$ are the positive measures on D with

$$\int_{D^2} G_D(x, y) \, d\mu(x) d\mu(y) < \infty$$

and $\mathcal{M}$ are the measures which are finite sums of these. The “value” of the Gaussian free field on this “blurred point” $\mu \in \mathcal{M}$ should then be

$$\int_D \psi(x) \, d\mu(x).$$

A *Gaussian free field* is a collection of random variables

$$\Gamma(\mu), \quad \mu \in \mathcal{M},$$

such that any linear sum of finitely many of these is Gaussian with mean zero, and

$$\mathbf{E}(\Gamma(\mu)\Gamma(\nu)) = \int_{D^2} G_D(x, y) \, d\mu(x) d\nu(y).$$

Proposition 1.3.1. A Gaussian free field exists and is unique.

Proof. Existence follows immediately from applying the Kolmogorov existence theorem. Indeed, for every finite subset $\mathcal{F} \subseteq \mathcal{M}$ we have a random variable valued

$$X_{\mathcal{F}} : \Omega \longrightarrow \mathbf{R}^{\mathcal{F}}$$

which is Gaussian of mean zero and covariance as above. This is well-defined because by a simple computation in [B], the bilinear form

$$(\mu, \nu) \longmapsto \int_{D^2} G_D(x, y) \, d\mu(x) d\nu(y)$$

on $\mathbf{R}^{\mathcal{F}}$ is positive definite. Since for any subset $\mathcal{F}' \subseteq \mathcal{F}$ these are compatible, $X_{\mathcal{F}}|_{\mathbf{R}^{\mathcal{F}'}} = X_{\mathcal{F}'}$, Kolmogorov says that these arise from a collection of random variables indexed by $\mu \in \mathcal{M}$:

$$X_{\mu} : \Omega \longrightarrow \mathbf{R}.$$

□

For instance, when $D = \mathbf{R}_+$, the Gaussian free field is simply Brownian motion.

1.4. Explicit construction. The Gaussian free field is a sum of independent Gaussians, each “spread out” over the space, as we will now see. To “spread out” each Gaussian, use ν_i , an orthonormal basis of $\mathcal{M}$,¹ then morally set the Gaussian free field to be

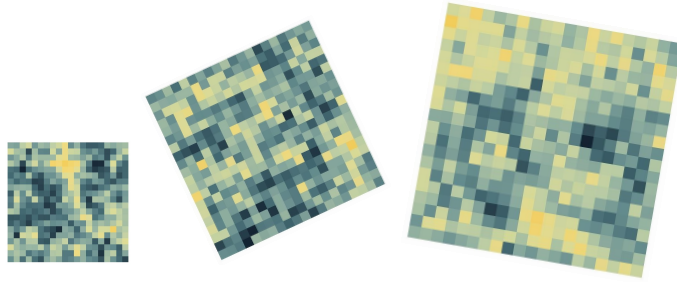
$$\psi = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i \nu_i.$$

Here $a_i \sim \mathcal{N}(0, 1)$ are independent Gaussians of mean zero and variance one. More precisely,

$$\Gamma(\mu) = \lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n a_i \nu_i, \mu \right\rangle.$$

So for instance $\Gamma(\nu_i) = a_i$, from which it follows that $\mathbf{E}(\Gamma(\mu)\Gamma(\nu)) = \langle \mu, \nu \rangle$, which proves that $\Gamma(\mu)$ is indeed a Gaussian free field.

1.5. Properties. When $d = 2$, something special happens: Gaussian free field is scale invariant, which follows because Green’s function is scale invariant too.



More generally, Green’s function is conformally invariant, and so the Gaussian free field is conformally invariant, meaning that

$$\Gamma_D(\mu) = \Gamma_{gD}(g_*\mu)$$

for every conformal transformation g .²

1.6. We can define the *correlation function* on D^n

$$(z_1, \dots, z_n) \mapsto \langle \Gamma(z_1)\Gamma(z_2) \cdots \Gamma(z_n) \rangle$$

to be a function such that for all measures μ_i ,

$$\mathbf{E}(\Gamma(\mu_1) \cdots \Gamma(\mu_n)) = \int_{D^n} \langle \Gamma(z_1)\Gamma(z_2) \cdots \Gamma(z_n) \rangle d\mu_1(z_1) \cdots d\mu_n(z_n).$$

Under certain conditions when $d = 2$, these can be shown to exist uniquely. Thus on the level of correlation functions, we did not lose any information in passing from ψ to Γ .

¹ $\mathcal{M}$ is a Hilbert space using the form $\langle \mu, \nu \rangle = \int_{D^2} G_D(x, y) d\mu(x) d\nu(y)$.

²Here $g_*\mu$ denotes the pushforward measure, defined by $g_*\mu(A) = \int_{g^{-1}A} |g'(x)|^2 d\mu(x)$.

1.7. There is a relation to the SLE_4 curve. To begin, one can define a Gaussian free field with boundary conditions. Then consider the unit disk D with boundary conditions $\pm\lambda$ on the upper and lower halves of ∂D . Thus the boundary condition has a “cliff-edge” at $+1$ and -1 . Given our Gaussian free field Γ , it is possible to make sense of its “cliff-edge”: a continuous simple curve γ from -1 to $+1$. Then for a certain value of λ , this curve is actually an SLE_4 curve! For more details, see [WE].

1.8. The putative relation to conformal field theory is this. There is the *affine* vertex algebra $\mathcal{A}_{\mathfrak{g},0}$ associated to any finite dimensional Lie algebra $\mathfrak{g}$. Then the Heisenberg algebra $\mathcal{A}_{\mathbf{C},0}$ is generated by a single field $b(z)$ which has genus zero correlation function

$$\langle 0|b(z)b(w)|0\rangle = \frac{1}{(z-w)^2}$$

which is (upon integrating) the same as the correlation function of the Gaussian free field. The multivariable Gaussian free field corresponds to abelian Lie algebras $\mathfrak{g} = \mathbf{C}^n$. To get it to work for general $\mathfrak{g}$, one possibility might be using noncommutative probability.

1.9. **Acknowledgements.** I would like to thank Dmitry Belyaev and Dylan Butson, who kindly explained many aspects of the Gaussian free field and factorisation homology to me, respectively.

2. CHIRAL ALGEBRAS

2.1. If the Gaussian free field is one way to mathematically formalise (part of) conformal field theory, then chiral and factorisation algebras as developed by Beilinson Drinfeld in [BD] give a completely different way.

2.2. Fix X a separated scheme of finite type over a field k of characteristic 0.

2.2.1. We want to model particles on X which can move around, and collide to form new particles. At very least this structure should be modelled by a $\mathcal{D}$ module

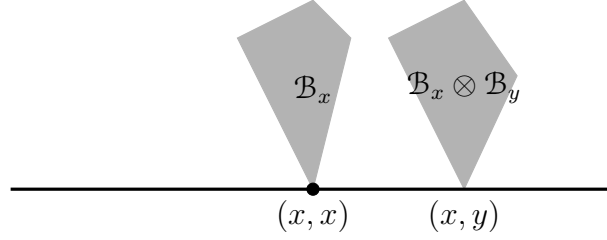
$$\mathcal{B} \in \mathcal{D}\text{-mod}(X)$$

whose fibre above a point $x \in X$ is the vector space $\mathcal{B}_x$ of possible particles living at x , and parallel transporting using the $\mathcal{D}$ module allows us to move them around.

To model *interactions*, we need to consider multiple particles living at points of X , using

$$\mathcal{B}_k \in \mathcal{D}\text{-mod}(X^k)$$

like before. Its fibre should be $\mathcal{B}_{x_1} \otimes \cdots \otimes \mathcal{B}_{x_n}$ when all the points x_i are different. More generally, if we allowing points to coincide the fibre is $\mathcal{B}_{\bar{x}_1} \otimes \cdots \otimes \mathcal{B}_{\bar{x}_r}$ where $\{\bar{x}_i\}$ are the distinct values of the x_i .



Pictured above are the fibres of $\mathcal{B}_2$. Thus, picking a pair of particles

$$\varphi \otimes \psi \in \mathcal{B}_x \otimes \mathcal{B}_y$$

living over distinct points, we can parallel transport this moving y along a path $y \rightarrow x$, giving a new element

$$\varphi \circ \psi \in \mathcal{B}_x$$

called the *fusion product*.

2.2.2. We first package the notion of a compatible family of $\mathcal{D}$ modules on X^k for each k . The object classifying finite subsets of X is

$$\mathrm{Ran}X = \mathrm{colim}_{I \in \mathrm{FSet}^{\mathrm{surj}}} X^I,$$

then the notion we are looking for is

$$\mathcal{D}\text{-mod}(\mathrm{Ran}X).$$

The above colimit is taken over all finite sets I in the category $\mathrm{FSet}^{\mathrm{surj}}$ of finite sets with surjections; any such surjection gives a diagonal map:

$$I \twoheadrightarrow J \quad \rightsquigarrow \quad \Delta_{I/J} : X^J \rightarrow X^I.$$

Although X^I are all schemes, the category of schemes is not closed under such colimits, and we need to take the above colimit in the category PreStk of *prestacks*. If Y is a prestack, its category of $\mathcal{D}$ -modules is defined as $\mathcal{D}\text{-mod}(Y) = \mathrm{QCoh}(Y_{dR})$, where quasicoherent sheaves on a prestack are defined by $\mathrm{QCoh}(\mathrm{Spec}A) = A\text{-mod}$ for a derived ring A and continuity of $\mathrm{QCoh}(-)$ under pullbacks.

Thus

$$\mathcal{D}\text{-mod}(\mathrm{Ran}X) = \mathrm{colim}_{I \in \mathrm{FSet}^{\mathrm{surj}}} \mathcal{D}\text{-mod}(X^I).$$

2.2.3. There is a symmetric monoidal structure on this category, given by

$$(\mathcal{A} \otimes^{ch} \mathcal{B})_I = \bigoplus_{I=I_1 \cup I_2, I_i \neq \emptyset} j_* j^*(\mathcal{A}_{I_1} \boxtimes \mathcal{B}_{I_2})$$

and more generally $(\otimes_{k \in K}^{ch} \mathcal{A}^k)_I = \bigoplus_{\varphi: I \rightarrow K} j_* j^* \boxtimes_{k \in K} \mathcal{A}_{\varphi^{-1}(k)}^k$, where j is the complementary open embedding to $\Delta_{I/K}$. It is called the *chiral* tensor product. Using it we can now formalise the notion that

$$\mathcal{B}_{(x,y)} = \begin{cases} \mathcal{B}_x \otimes \mathcal{B}_y & \text{if } x \neq y, \\ \mathcal{B}_x & \text{if } x = y. \end{cases}$$

This can be rephrased as

$$j^* \mathcal{B}_{\{1,2\}} = j^* (\mathcal{B} \otimes^{ch} \mathcal{B})_{\{1,2\}}.$$

Definition 2.2.4. Take a cocommutative coassociative coalgebra $\mathcal{B} \in (\mathcal{D}(\text{Ran} X), \otimes^{ch})$. Iterating the comultiplication map gives

$$\mathcal{B} \rightarrow \otimes_{k \in K}^{ch} \mathcal{B}.$$

In particular, for every surjection $\varphi: I \twoheadrightarrow K$ we get

$$\mathcal{B}_I \rightarrow j_* j^* \boxtimes_{k \in K} \mathcal{B}_{\varphi^{-1}(k)}.$$

$\mathcal{B}$ is called a **factorisation algebra** if each $j^* \mathcal{B}_I \rightarrow j^* \boxtimes_{k \in K} \mathcal{B}_{\varphi^{-1}(k)}$ is an equivalence.

2.3. Vertex algebras. Assume that X is smooth, and let d denote its dimension. If $\mathcal{B} \in \mathcal{D}\text{-mod}(\text{Ran} X)$, we can consider the Mayer Vietoris sequence of $\mathcal{D}$ -modules on X^2 :

$$\Delta_* \Delta^! \mathcal{B}_{\{1,2\}} \rightarrow \mathcal{B}_{\{1,2\}} \rightarrow j_* j^* \mathcal{B}_{\{1,2\}} \xrightarrow{+1}$$

Note that $\Delta^! \mathcal{B}_{\{1,2\}} = \Delta^* \mathcal{B}_{\{1,2\}}[-d] = \mathcal{B}_{\{1\}}[-d]$.

If $\mathcal{B}$ is a factorisation algebra, the open part can be simplified, giving

$$\mathcal{B}_{\{1,2\}} \rightarrow j_*(\mathcal{B}_{\{1\}} \boxtimes \mathcal{B}_{\{2\}}) \rightarrow \Delta_* \mathcal{B}_{\{1\}}[-d+1] \xrightarrow{+1}$$

If X is a curve and $\mathcal{B}$ is concentrated in degree zero, then we get a short exact sequence in $\mathcal{D}\text{-mod}(X^2)^\heartsuit$

$$0 \rightarrow \mathcal{B}_{\{1,2\}} \rightarrow j_*(\mathcal{B}_{\{1\}} \boxtimes \mathcal{B}_{\{2\}}) \xrightarrow{Y} \Delta_* \mathcal{B}_{\{1\}} \rightarrow 0$$

When restricted to the product of two formal disks at x , the above Y is precisely the vertex algebra map. See [FB, §20] for more details.

2.4. Factorisation spaces. There is a space version of factorisation algebras. If $\mathcal{S}, \mathcal{T}$ are prestacks living over the Ran space, we can take the pullback

$$\begin{array}{ccc} \mathcal{S} \times^{ch} \mathcal{T} & \longrightarrow & \mathcal{S} \times \mathcal{T} \\ \downarrow & & \downarrow \\ (\text{Ran} X \times \text{Ran} X)_{\text{disj}} & \longrightarrow & \text{Ran} X \times \text{Ran} X \end{array}$$

This gives the category $\mathrm{Sp}_X = \mathrm{PreStk}_{/(\mathrm{Ran}X)}$ a symmetric monoidal structure, called the *chiral product*. Explicitly, if I is a finite set then

$$(\mathcal{S} \times^{\mathrm{ch}} \mathcal{T})_I = \coprod_{I=I_1 \cup I_2, I_i \neq \emptyset} j_* j^*(\mathcal{S}_{I_1} \times \mathcal{T}_{I_2})$$

where $j : U \rightarrow X^I$ is the open embedding corresponding to I_1, I_2 .

2.4.1. If $f : U \rightarrow V$ is a map of prestacks, we have the adjunction (f_*, f^*) :

$$f_* : \mathrm{PreStk}_U \rightleftarrows \mathrm{PreStk}_V : f^*$$

where f^* is given by pullback and f_* is the forgetful functor.

Definition 2.4.2. A **factorisation space** $\mathcal{S}$ is a commutative associative algebra in the symmetric monoidal category $(\mathrm{Sp}_X, \times^{\mathrm{ch}})$, such that for every surjection $\varphi : I \twoheadrightarrow K$, the map

$$j^* \prod_{k \in K} \mathcal{S}_{\varphi^{-1}(k)} \longrightarrow j^* \mathcal{S}_I$$

is an equivalence, where $j : U \rightarrow X^I$ is the open embedding corresponding to φ .

2.4.3. For example, $\mathrm{Ran}X$ itself is a factorisation space. There is a correspondence

$$\begin{array}{ccc} & (\mathrm{Ran}X \times \mathrm{Ran}X)_{\mathrm{disj}} & \\ \swarrow & & \searrow \\ \mathrm{Ran}X \times \mathrm{Ran}X & & \mathrm{Ran}X \end{array}$$

which also makes $\mathrm{Ran}X$ into a commutative monoid in the category $\mathrm{Corr}(\mathrm{PreStk})$ of prestacks with morphisms given by correspondences. The same is true for any factorisation space $\mathcal{S}$.

2.4.4. Exactly the same definitions may be made for $\mathrm{Sp}_{X,dR} = \mathrm{PreStk}_{/(\mathrm{Ran}X_{dR})}$.

2.4.5. A *unit* of a factorisation space is a section

$$e : \mathrm{Ran}X_{dR} \longrightarrow \mathcal{S}$$

of the structure map $p : \mathcal{S} \rightarrow \mathrm{Ran}X_{dR}$. Applying IndCoh gives

$$\Gamma_{\mathcal{S}} : \mathcal{D}\text{-mod}(\mathrm{Ran}X) \xrightarrow{e_*} \mathcal{D}\text{-mod}(\mathcal{S}) \xrightarrow{\mathrm{Oblv}} \mathrm{IndCoh}(\mathcal{S}) \xrightarrow{p^*} \mathcal{D}\text{-mod}(\mathrm{Ran}X)$$

Proposition 2.4.6. For every factorisation space with unit $(\mathcal{S}, e)$, the $\mathcal{D}$ -module $\Gamma_{\mathcal{S}}(\omega)$ is a factorisation algebra.

There is a twisted version of the above. Let $\mathcal{L}$ be a factorisation line bundle on $\mathcal{S}$, i.e. for every surjection $\varphi : I \twoheadrightarrow K$ there are compatible isomorphisms

$$j^* \mathcal{L}_I \longrightarrow j^* \otimes_{k \in K} \mathcal{L}_{\varphi^{-1}(k)}.$$

Then we have

$$\Gamma_{\mathcal{S}, \mathcal{L}} : \mathcal{D}\text{-mod}(\text{Ran}X) \xrightarrow{e_*} \mathcal{D}\text{-mod}(\mathcal{S}) \xrightarrow{\text{Oblv}} \text{IndCoh}(\mathcal{S}) \xrightarrow{\otimes^{\mathcal{L}}} \text{IndCoh}(\mathcal{S}) \xrightarrow{p_*} \mathcal{D}\text{-mod}(\text{Ran}X)$$

and similarly

Proposition 2.4.7. For every factorisation space with unit $(\mathcal{S}, e)$, and every factorisation line bundle $\mathcal{L}$, the $\mathcal{D}$ -module $\Gamma_{\mathcal{S}, \mathcal{L}}(\omega)$ is a factorisation algebra.

For details, see for instance [FB, §20].

2.5. Main example. Let X be a curve and G a reductive group over k . If $x_1, \dots, x_n$ are points on X , then consider

$$\text{Gr}_{G, x_1, \dots, x_n}(k) = \{P \in \text{Bun}_G(X), \varphi : P|_{X \setminus \{x_i\}} \simeq \text{triv}|_{X \setminus \{x_i\}}\}.$$



This classifies G bundles on the curve with “singularities” (otherwise known as *Hecke modifications*) concentrated at the points $x_1, \dots, x_n$. Note that every G bundle is trivial away from finitely many points.

This arrangement gives a prestack called the *Beilinson-Drinfeld Grassmannian*

$$\text{Gr}_{G, \text{Ran}X} \longrightarrow \text{Ran}X$$

which classifies a G bundle along with a trivialisation away from a finite subset $\{x_i\} \subset X$. This map is ind-schematic, so that the fibres $\text{Gr}_{X, x_1, \dots, x_n}$ are ind-schemes.

2.5.1. To be precise, the value of its functor of points at a k -algebra A is

$$\text{Gr}_{G, \text{Ran}X}(A) = \{\{x_i\} \subset X(A), P \in \text{Bun}_G(X_A), \varphi : P|_{X \setminus \{x_i\}} \simeq \text{triv}|_{X \setminus \{x_i\}}\}$$

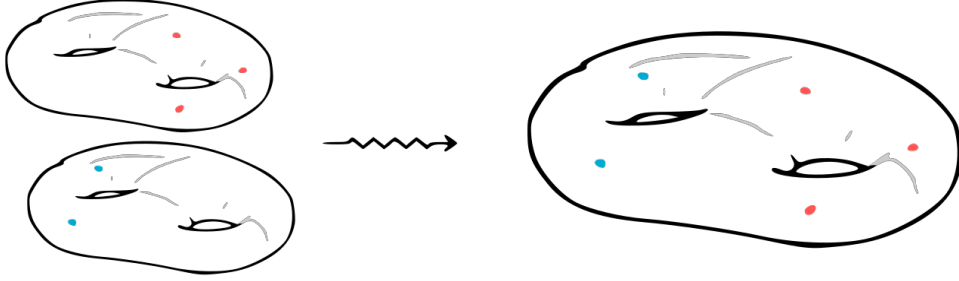
where $\text{Bun}_G(X_A)$ is the groupoid of $\underline{G}$ torsors on the base change X_A .

2.5.2. Notice first that if the points $x_1, \dots, x_n$ are distinct,

$$(1) \quad \text{Gr}_{G, x_1, \dots, x_n} \simeq \text{Gr}_{X, x_1} \times \cdots \times \text{Gr}_{X, x_n}.$$

This follows from the Beauville-Laszlo theorem, which implies that a G bundle on X is uniquely determined by its restriction to $X \setminus \{x_1, \dots, x_n\}$, to the formal disks D_{x_i} , and by the gluing data on the punctured formal disks $D_{x_i}^\times = D_{x_i} \cap (X \setminus \{x_1, \dots, x_n\})$.

2.5.3. The identification (1) can be extended to the structure of a factorisation space on $\mathrm{Gr} = \mathrm{Gr}_{X, \mathrm{Ran}X}$.



Indeed, note that to give map

$$(\mathrm{Gr} \times^{\mathrm{ch}} \mathrm{Gr}) = \coprod_{\varphi: I \twoheadrightarrow \{1,2\}} j_* j^*(\mathrm{Gr} \times \mathrm{Gr}) \longrightarrow \mathrm{Gr}$$

by adjunction is the same thing as to give for each φ a map

$$j^*(\mathrm{Gr} \times \mathrm{Gr}) \longrightarrow j^* \mathrm{Gr}.$$

The left hand side classifies disjoint subsets $\{x_{i_1} : i_1 \in I_1\}, \{x_{i_2} : i_2 \in I_2\} \subset X$, G bundles P_1, P_2 and trivialisations away from these subsets. Restricting the trivialisations to $X \setminus \{x_{i_1}\} \cup \{x_{i_2}\}$ and using them to glue P_1, P_2 gives a new G bundle with a trivialisation away from $\{x_{i_1}\} \cup \{x_{i_2}\}$. Associativity and commutativity is easy to check.

2.5.4. Picking a coordinate t around a point x , we have

$$\mathcal{O} = \mathcal{O}(D_x) \simeq k[[t]], \quad \mathcal{K} = \mathcal{O}(D_x^\times) \simeq k((t))$$

and then Beauville-Laszlo implies that

$$\mathrm{Gr}_{G,x}(k) = G(\mathcal{K})/G(\mathcal{O})$$

which is called the *affine Grassmannian*.

2.5.5. The Beilinson-Drinfeld Grassmannian carries a unit, coming from the trivial G bundle

$$e : \mathrm{Ran}X \longrightarrow \mathrm{Gr}_{G, \mathrm{Ran}X}.$$

Let $\mathfrak{g}$ be semisimple. The *affine factorisation algebra* at level zero is

$$\mathcal{A}_{\mathfrak{g},0} = \Gamma_{\mathrm{Gr}_{G, \mathrm{Ran}X}}(\omega)$$

where G is the simply connected algebraic group with Lie algebra $\mathfrak{g}$. In this case,

$$\mathrm{Pic}(\mathrm{Bun}_G(X)) = \mathbf{Z} \cdot \mathcal{L}$$

is generated by the so-called *theta* line bundle $\mathcal{L}$. We also denote its pullback to $\mathrm{Gr}_{G, \mathrm{Ran}X}$ by $\mathcal{L}$, and call it the *determinant* line bundle. Applying the twisted construction gives the *affine factorisation algebra* at level k :

$$\mathcal{A}_{\mathfrak{g}, k} = \Gamma_{\mathrm{Gr}_{G, \mathrm{Ran}X}, \mathcal{L}^{\otimes k}}(\omega).$$

The associated vertex algebra is called the *affine vertex algebra* at level k , and can be defined for any $k \in \mathbf{C}$ and any finite dimensional Lie algebra $\mathfrak{g}$ with an ad-invariant inner product κ (in the above it was the Killing form).

2.5.6. Applying this to $G = \mathbf{G}_m$ gives the simplest example, the *Heisenberg* vertex algebra. It is generated by a single field $b(z)$, satisfying

$$b(z)b(w) = \frac{\mathrm{id}}{(z-w)^2} + \text{power series in } (z-w).$$

The affine vertex algebra at level k is generated by fields $\alpha(z)$ for $\alpha \in \mathfrak{g}$, subject to

$$\alpha(z)\beta(w) = k \frac{\kappa(\alpha, \beta)}{(z-w)^2} + \frac{[\alpha, \beta](w)}{z-w} + \text{power series in } (z-w).$$

2.6. Correlation functions and factorisation homology. Let $\mathcal{B}$ be a factorisation algebra. Its *factorisation homology* is

$$H_{\mathrm{ch}}(X, \mathcal{B}) = H_{\mathrm{dR}}(\mathrm{Ran}X, \mathcal{B}) = p_* \mathcal{B},$$

meaning its $\mathcal{D}$ -module pushforward via $p : \mathrm{Ran}X \rightarrow \mathrm{pt}$. It is a coassociative cocommutative coalgebra. Similarly, its *factorisation homology* is

$$H^{\mathrm{ch}}(X, \mathcal{B}) = p_! \mathcal{B}.$$

If X is proper, then so is $\mathrm{Ran}X$, i.e. $p_* = p_!$.

2.6.1. In particular, by adjunction there is a map

$$\mathcal{B} \longrightarrow p^! p_! \mathcal{B} = \omega_{\mathrm{Ran}X} \otimes H^{\mathrm{ch}}(X, \mathcal{B}).$$

Thus factorisation homology is the “universal recipient of correlators”. Explicitly, what we mean by that is this. Pick a *conformal block*, which is an element

$$\psi \in H^{\mathrm{ch}}(X, \mathcal{B})^\vee.$$

Some sources require that it lives in H^0 .

Now, let b_i be flat sections of $\mathcal{B}_{\{1\}} \in \mathcal{D}\text{-mod}(X)$ on formal disks D_{x_i} about x_i . Since

$$j^* \mathcal{B}_{\{1, \dots, n\}} \simeq j^* (\mathcal{B}_{\{1\}} \boxtimes \cdots \boxtimes \mathcal{B}_{\{1\}})$$

on the complement $j : U \hookrightarrow X^n$ of all the diagonals, we get a flat section

$$b_1 \otimes \cdots \otimes b_n \in H_{\mathrm{dR}}^0(D_x^\times, \mathcal{B})$$

on $D_x^\times = U \cap (D_{x_1} \times \cdots \times D_{x_n})$. Applying the conformal block gives the composite map of $\mathcal{D}$ -modules

$$\mathcal{B}|_{D_x^\times} \longrightarrow \omega_{D_x^\times} \otimes H^\text{ch}(X, \mathcal{B}) \xrightarrow{\psi} \omega_{D_x^\times}.$$

We can apply H_{dR}^0 to get a map on the level of de Rham cohomology. The differential form

$$\psi(b_1 \otimes \cdots \otimes b_n) \in H_{\text{dR}}^0(D_x^\times, \omega)$$

is called an *n-point correlation function*.

2.6.2. For instance, when $X = \mathbf{P}^1$ one can show that $H_0^\text{ch}(X, \mathcal{B}) = \mathbf{C}\langle 0|$ is one dimensional, generated by the functional which picks out the component of the vacuum $|0\rangle$. Then taking the flat sections $b(z)|0\rangle$ and $b(w)|0\rangle$ gives two-point correlation function

$$\langle 0|b(z)b(w)|0\rangle = \frac{1}{(z-w)^2}.$$

3. DISCRETE APPROXIMATIONS

3.1. The *discrete Gaussian free field* is a collection of random variables

$$\psi(x)$$

labelled by points of a subset $D \subseteq \mathbf{Z}^d$, which are each Gaussian with mean zero and have correlations

$$\mathbf{E}[\psi(x)\psi(y)] = G_D(x, y).$$

Here G_D denotes the Green's function for minus the discrete Laplacian $\Delta f(x) = \frac{1}{2d} \sum_{x' \sim x} f(x') - f(x)$. It is not difficult to show that

Lemma 3.1.1. $G_D(x, y) = \sum_{k \geq 0} \#\{\text{length } k \text{ paths } x \rightarrow y \text{ in } D\} \left(\frac{1}{2d}\right)^k$.

Proof. Verify that $\Delta_x G_D(x, y) = -\delta_{x, y}$. □

Note that the Gaussian free field can be viewed as a random variable taking values in $\mathbf{R}^D$. One can show that it has density function proportional to

$$(2) \quad \rho : \mathbf{R}^D \longrightarrow \mathbf{R} \quad (\gamma_x) \longmapsto \exp\left(-\frac{1}{2 \cdot 2d} \sum_{x \text{ or } y \in D} |\gamma_x - \gamma_y|^2\right)$$

where we have set $\gamma_x = 0$ if x lies outside D .

Proposition 3.1.2. The above density (2) is a discrete Gaussian free field.

Proof. Fix an $x \in D$. Conditional on $\psi(y) = h(y)$ for all $y \neq x$,

$$\psi(x) \sim \mathcal{N}(1, \bar{h}(x)),$$

where $\bar{h}(y) := \frac{1}{2d} \sum_{x' \sim y} h(x')$. In particular, $\psi(x)$ only depends on the average of its neighbours. Thus if $x \neq y$,

$$\mathbf{E}[\psi(x)\psi(y)] = \mathbf{E}[\psi(x)\bar{\psi}(y)] = \frac{1}{2d} \sum_{y' \sim y} \mathbf{E}[\psi(x)\psi(y')]$$

and similarly

$$\begin{aligned} \mathbf{E}[\psi(x)\psi(x)] &= \mathbf{E}[\psi(x)\bar{\psi}(x)] + \mathbf{E}[(\psi(x) - \bar{\psi}(x))^2] + \mathbf{E}[(\psi(x) - \bar{\psi}(x))\bar{\psi}(x)] \\ &= \frac{1}{2d} \sum_{x' \sim x} \mathbf{E}[\psi(x)\psi(x')] + 1 + 0. \end{aligned}$$

Unwinding the definitions, this says precisely that $\Delta_x \mathbf{E}[\psi(x)\psi(y)] = -\delta_{x,y}$. $\square$

3.1.3. We can make the same definition for any countable set of vertices with weighted arrows between each, such that the sum of the weights $\rho_{x,y}$ flowing out of any particular vertex x is 1. The density is now

$$(\gamma_x) \longmapsto \exp \left(-\frac{1}{2} \sum_{x \text{ or } y \in D} \rho_{x,y} |\gamma_x - \gamma_y|^2 \right)$$

the finite Laplacian is

$$\Delta f(x) = \sum_{x \sim y} \rho_{x,y} (f(y) - f(x))$$

and its Green's function is

$$G_D(x, y) = \sum_{k \geq 0} \mathbf{P}(\text{a length } k \text{ walk starting at } x \text{ is at } y)$$

where the Markov random walk is given by probabilities $\rho_{x,y}$.

3.2. **Physics remarks.** One should view $\mathbf{R}^D$ as being the set of fields on D , and a field (γ_x) has energy

$$H(\gamma) = \frac{1}{2d} \sum_{x \text{ or } y \in D} |\gamma_x - \gamma_y|^2.$$

In statistical field theory, in such a situation the probability of a field γ is

$$P(\gamma, \beta) = \frac{e^{-\beta H(\gamma)}}{\int e^{-\beta H(\gamma)} d\gamma}$$

for a parameter $\beta > 0$ called the *temperature*. In particular, one should view the Gaussian free field as being an analogue of the Ising model (where $\psi(x)$ takes values in $\mathbf{Z}/2$) and the XY model (where $\psi(x)$ takes values in S^1).

3.2.1. One can show, e.g. as in [GS], that something like the Gaussian free field arises as the critical temperature limit of other statistical field theories, e.g. the XY model.

APPENDIX A. GREEN'S FUNCTIONS

A.1. The idea behind Green's functions is to solve differential equations by “inverting” differential operators

$$\Delta\varphi(x) = f(x) \iff \varphi(x) = \int f(y)G(x,y)dy.$$

An inverse G is called a *Green's function* of Δ , and viewing G as a distribution, it can be computed by solving

$$\Delta_x G(x,y) = \delta(x-y).$$

A.1.1. For instance, the function $G(z,w) = \log|z-w|$ is a Green's function for the Laplacian on the complex plane $\mathbf{C}$.

A.1.2. When the spectrum of the Laplacian is discrete, its Green's function can be explicitly written in terms of an orthonormal basis of eigenfunctions φ_i with eigenvalue λ_i :

$$G(x,y) = \sum \frac{1}{\lambda_i} \varphi_i(x) \varphi_i(y).$$

More generally, allowing the spectrum is continuous, we can compute it using Fourier transforms. The Fourier transform of $\delta(x-y)$ is $e^{ik \cdot y}$ and the Fourier transform of Δ_x is $|k|^2$, so using Fourier inversion we have

$$G(x,y) = \int \frac{e^{ik \cdot y}}{|k|^2} dk.$$

A.2. Physically, if we consider an electrically charged body whose density at point x is $\rho(x)$, then its electrostatic potential is

$$\Delta^{-1}\rho(x) = - \int_D G_D(x,y)\rho(y)dy$$

and the covariance

$$\int_{D \times D} G_D(x,y)\rho(x)\rho(y)dxdy$$

is the energy it took to overcome the repulsive electric forces, and bring the charges in from far away from each other to their current position.

REFERENCES

- [B] Berestycki, N., 2015. *Introduction to the Gaussian free field and Liouville quantum gravity*. Lecture notes.
- [BD] Beilinson, A., Drinfeld, V. and Drinfeld, V.G., 2004. *Chiral algebras* (Vol. 51). American Mathematical Soc..
- [FB] Frenkel, E. and Ben-Zvi, D., 2004. *Vertex algebras and algebraic curves* (No. 88). American Mathematical Soc..
- [FG] Francis, J. and Gaitsgory, D., 2012. *Chiral koszul duality*. Selecta Mathematica, 18(1), pp.27-87.
- [G] Gaitsgory, D., 2011. *Ind-coherent sheaves*. arXiv preprint. <https://arxiv.org/pdf/1105.4857.pdf>
- [GS] Garban, C. and Sepúlveda, A., 2020. Statistical reconstruction of the Gaussian free field and KT transition. arXiv preprint. <https://arxiv.org/abs/2002.12284>
- [WE] Werner, W. and Powell, E., 2020. *Lecture notes on the Gaussian free field*. arXiv preprint. arxiv.org/abs/2004.04720